Topic 7: Nuclear and Particle Physics

Topic 7A: The Nuclear Atom and Atomic Structure

1. The Nuclear Atom

Before the discovery of the atomic nucleus, the prevailing model of the atom was J.J. Thomson's **"Plum Pudding" model**. Thomson envisioned the atom as a sphere of uniform positive charge with negatively charged electrons embedded within it, like raisins in a pudding.

Rutherford's Alpha Particle Scattering Experiment

In 1909, Ernest Rutherford, along with his colleagues Hans Geiger and Ernest Marsden, performed a groundbreaking experiment that completely overturned Thomson's model.

- **The Experimental Setup:**
  - A source of alpha particles (α-particles, which are helium nuclei consisting of 2 protons and 2 neutrons) was placed inside a lead collimator to produce a narrow beam.
  - This beam was directed at an extremely thin sheet of gold foil (only a few hundred atoms thick).
  - A movable zinc sulfide (ZnS) scintillation screen attached to a microscope was placed around the gold foil inside a vacuum chamber to detect scattered alpha particles. When an alpha particle struck the screen, it produced a tiny flash of light (a scintillation).
  - A vacuum was required because alpha particles have a short range in air (only a few centimeters) and would easily collide with air molecules, ruining the accuracy of the experiment.
- **Observations and Conclusions:**
  The results were highly unexpected based on the Plum Pudding model, which predicted that the weak, spread-out positive charge of the atom would only slightly deflect the fast-moving, massive alpha particles.

Observation	Conclusion
1. The vast majority of alpha particles passed straight through the gold foil with virtually no deflection.	The atom is mostly **empty space**.
2. A small fraction of alpha particles were deflected through small to moderate angles.	There is a concentration of **positive charge** within the atom that repels the positively charged alpha particles via electrostatic forces.

Observation	Conclusion
3. An extremely small number (about 1 **in** 8000**) were deflected by more than** $90°$**, sometimes bouncing straight back.**	The positive charge and almost **all the mass** of the atom must be concentrated in a tiny, dense region at the center called the **nucleus**.

Distance of Closest Approach

When an alpha particle travels directly toward a gold nucleus, it experiences a repulsive electrostatic force that slows it down. At the point of closest approach (d), its kinetic energy (E_k) is momentarily completely converted into electrostatic potential energy (E_p).

This principle allows physicists to estimate an upper limit for the radius of a nucleus:

$$E_k = \frac{Q_1 Q_2}{4\pi\varepsilon_0 d}$$

Where:

- $Q_1 = 2e$ (charge of the alpha particle)
- $Q_2 = Ze$ (charge of the target gold nucleus, where Z is the atomic number)
- ε_0 is the permittivity of free space

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2. Electrons from Atoms

Electrons orbit the dense, central positive nucleus at relatively immense distances. To probe atomic and nuclear structures more deeply than alpha particles allow, physicists use thermionic emission and high-energy electron beams.

Thermionic Emission

When a metal filament is heated by an electric current, the free conduction electrons gain kinetic energy. If they gain sufficient thermal energy to overcome the attractive electrostatic forces holding them to the metal surface (overcoming the metal's **work function**), they escape the surface. This process is called **thermionic emission**.

Accelerating and Deflecting Electrons

Once emitted, these electrons can be accelerated to high speeds using an electric field. By placing a positively charged anode opposite the heated cathode, the electrons experience an accelerating force.

The work done on an electron by a potential difference V is given by:

$$W = eV$$

Assuming the electron starts from rest, this work becomes its kinetic energy:

$$eV = \frac{1}{2}mv^2$$

Where:

- e is the elementary charge (1.60×10^{-19} C)
- m is the mass of the electron
- v is the final velocity of the electron

Once accelerated into a beam, these electrons can be deflected by:

1. **Electric Fields:** Exerting a constant force parallel to the field lines ($F = EQ$), causing a parabolic trajectory.
2. **Magnetic Fields:** Exerting a magnetic force perpendicular to the velocity ($F = BQv$), forcing the electrons into a circular path.

Topic 7B: Particle Accelerators and Particle Detectors

To explore the fundamental building blocks of matter, particles must be accelerated to extreme kinetic energies. High energies are necessary for two main reasons:

1. **To overcome strong electrostatic repulsion** between like-charged particles.
2. **To resolve smaller structures.** According to the de Broglie hypothesis, higher momentum (p) yields a shorter wavelength ($\lambda = h/p$). Finer details require shorter wavelengths.

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1. Particle Accelerators

Common Operational Principles

Based on the experimental standard, both linear accelerators (linacs) and cyclotrons share two core structural and functional traits:

- Both types of accelerators operate using a **fixed frequency alternating potential difference (p.d.)**.
- Both types of accelerators utilize **electric fields to accelerate** the charged particles.

Linear Accelerators (Linacs)

- **Structure & Working:**
  - Linacs consist of a series of metal drift tubes of **increasing length** aligned in a straight line.
  - An **electric field** is created across the gaps between these tubes by the alternating p.d. source. It is this **electric field** that accelerates the charged particles whenever they cross a gap.

- Inside the hollow metal drift tubes, the **electric field is zero** (acting as a Faraday cage), meaning particles drift through them at a constant speed.
- The tubes must get progressively longer because the **particles travel through each successive tube with a higher speed**. To ensure the particles always arrive at the next gap exactly when the fixed-frequency alternating electric field switches direction to accelerate them forward, they must spend an equal amount of time inside every tube.

Cyclotrons

- **Structure & Working:**
  - A cyclotron uses **circular, D-shaped electrodes** (called Dees) separated by a small gap, enclosed in a vacuum.
  - A **magnetic field perpendicular to the motion** of the particles is applied across the Dees. This magnetic field exerts a centripetal force that deflects the particles into a circular path, but it does *not* change their speed.
  - The **electric field** exists purely in the narrow gap between the Dees, established by a **fixed frequency alternating p.d.**
  - As particles cross the gap, the **electric field accelerates them**, increasing their speed and momentum (p).
  - The alternating voltage is timed to reverse its polarity precisely while the particle is inside the shielding of a Dee, ensuring that every time a particle enters the gap, the **electric field** is oriented in the correct direction to accelerate it.
  - As the particle's speed increases across the gap, its orbital radius within the next Dee must increase according to the equation:

    $$r = \frac{p}{BQ}$$

    Because $p = mv$, **the radius of the circle travelled by the particle increases with speed**, causing it to follow a spiral path until it exits the machine.
  - The fixed frequency (f) of the alternating voltage matching the circular path of the particle is given by:

    $$f = \frac{BQ}{2\pi m}$$

The Large Hadron Collider (LHC)

The LHC at CERN is a circular synchrotron accelerator where particles are constrained to a fixed ring path.

- As the particles gain momentum (p), the strength of the magnetic field (B) is continuously increased to maintain a constant orbital radius ($r = \frac{p}{BQ}$).
- Radiofrequency (RF) cavities along the ring provide the electric fields that accelerate the counter-rotating proton beams close to the speed of light before forcing them to collide.

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2. Particle Detectors

Detectors track and identify particles produced in high-energy collisions by relying on moving particles ionizing atoms along their path.

Track Detection in Magnetic Fields

Detectors are placed within powerful uniform magnetic fields. When a moving charged particle enters this field, the magnetic force acts as a centripetal force, allowing its properties to be mapped through its path radius:

$$r = \frac{p}{BQ}$$

- **Charge Sign ($+/-$):** The direction of curvature (clockwise vs. counter-clockwise) reveals the sign of the charge via Fleming's Left-Hand Rule.
- **Momentum (p):** A larger radius of curvature (r) corresponds to a larger momentum (a straighter track). A smaller radius means lower momentum (a tightly curved track).
- **Energy Loss:** As particles ionize the medium, they lose kinetic energy and momentum, causing the path radius to shrink continuously into a **spiral**.

Examples of Historical and Modern Detectors

- **Bubble Chambers:** Filled with a superheated liquid where ionizing particles leave visible trails of boiling bubbles.
- **Cloud Chambers:** Contain a supersaturated vapor where ionizing particles leave paths of condensation droplets.
- **Modern Electronic Detectors (e.g., ATLAS, CMS at LHC):** Use layers of silicon pixel detectors and wire drift chambers to digitally record track coordinates and particle momentum instantly.

Topic 7C: Particle Interactions, Reactions, and the Standard Model

1. Mass-Energy Equivalence

During high-energy particle collisions, kinetic energy can be converted directly into the mass of entirely new particles according to Einstein's mass-energy equation:

$$\Delta E = c^2 \Delta m$$

Where:

- ΔE is the change in energy
- Δm is the change in mass
- c is the speed of light in a vacuum ($\approx 3.00 \times 10^8$ m s^{-1})

Pair Production and Annihilation

- **Pair Production:** A high-energy photon transforms into a particle and its corresponding antiparticle ($\gamma \rightarrow e^- + e^+$). The photon's energy ΔE must be at least equal to the combined rest mass energy of both particles.
- **Annihilation:** When a particle meets its corresponding antiparticle, they destroy one another, converting their entire mass into energy released as a pair of photons.

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2. The Particle Zoo and The Standard Model

Every known subatomic particle fits into a classification scheme known as the **Standard Model**.

Matter is divided into two primary categories based on whether they feel the Strong Nuclear Force: **Leptons** and **Hadrons**.

Leptons (Fundamental Particles)

Leptons are fundamental particles that **do not** experience the Strong Nuclear Force. They interact via the Weak Nuclear Force, Electromagnetic Force (if charged), and Gravity.

The lepton family includes:

1. **Electron (e^-)** and **Electron Neutrino (ν_e)**
2. **Muon (μ^-)** and **Muon Neutrino (ν_μ)**
3. **Tau (τ^-)** and **Tau Neutrino (ν_τ)**

- Charged leptons have a charge of $-1e$. Neutrinos have a charge of 0.
- Every lepton has an **Antilepton** counterpart (e.g., the positron e^+, and the electron antineutrino $\bar{\nu}_e$).

Hadrons (Composite Particles)

Hadrons are composite particles made of smaller fundamental components called **Quarks**. Because they contain quarks, hadrons experience the **Strong Nuclear Force**.

Hadrons are subdivided into two classes:

1. **Baryons:** Made of **three quarks** (qqq). Examples include:
   - **Proton (p):** Quark composition is uud.
   - **Neutron (n):** Quark composition is udd.
2. **Mesons:** Made of **one quark and one antiquark pair** $(q\bar{q})$. Examples include:
   - **Pions (π-mesons):** e.g., $\pi^+(u\bar{d}), \pi^-(d\bar{u})$.
   - **Kaons (K-mesons):** Contain a strange (s) or antistrange $(\bar{s})$ quark.

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3. Quarks

Quarks are fundamental particles that carry fractional electric charges. They exist in six flavors:

Quark Flavor	Symbol	Charge (e)	Baryon Number (B)	Strangeness (S)
Up	u	$+\frac{2}{3}$	$+\frac{1}{3}$	0
Down	d	$-\frac{1}{3}$	$+\frac{1}{3}$	0
Charm	c	$+\frac{2}{3}$	$+\frac{1}{3}$	0
Strange	s	$-\frac{1}{3}$	$+\frac{1}{3}$	-1
Top	t	$+\frac{2}{3}$	$+\frac{1}{3}$	0
Bottom	b	$-\frac{1}{3}$	$+\frac{1}{3}$	0

- **Antiquarks** $(\bar{u}, \bar{d}$, etc.) carry **exactly opposite signs** for Charge, Baryon Number, and Strangeness.
- **Quark Confinement:** Quarks can never be isolated individually. Attempting to pull them apart injects energy into the gluon field, creating a new quark-antiquark pair via $\Delta E = c^2 \Delta m$ rather than freeing a single quark.

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4. Particle Reactions and Conservation Laws

In any nuclear or particle reaction, several quantities must be **strictly conserved**. If a proposed reaction violates any of these conservation laws, the reaction is physically impossible.

Summary of Conservation Laws

1. **Charge (Q):** Total electric charge before the reaction must equal total electric charge after.
2. **Baryon Number (B):** Quarks have $B = +\frac{1}{3}$; antiquarks have $B = -\frac{1}{3}$. Total baryon number must be conserved.
3. **Lepton Numbers (L_e, L_μ, L_τ):** Lepton numbers must be conserved **by generation**. For example, total electron-lepton number (L_e) must balance separately from muon-lepton number (L_μ).
4. **Mass-Energy and Momentum:** Total energy (including rest mass energy) and momentum must be conserved.

Special Rule for Strangeness (S):
* Strangeness is **conserved** during interactions governed by the **Strong Nuclear Force** and **Electromagnetic Force**.
* Strangeness can be **violated by** ± 1 during interactions governed by the **Weak Nuclear Force** (such as weak decays involving quark flavor changes).

PARTICLE	SYMBOL	CHARGE	LEPTON NUMBER, L	BARYON NUMBER, B	ANTI-PARTICLE	SYMBOL	CHARGE	LEPTON NUMBER, L	BARYON NUMBER, B
electron	e^-	-1	1	0	positron	e^+	+1	-1	0
electron neutrino	ν_e	0	1	0	anti-electron neutrino	$\bar{\nu}_e$	0	-1	0
muon	μ	-1	1	0	anti-muon	$\bar{\mu}$	+1	-1	0
muon neutrino	ν_μ	0	1	0	anti-muon neutrino	$\bar{\nu}_\mu$	0	-1	0
tau	τ	-1	1	0	anti-tau	$\bar{\tau}$	+1	-1	0
tau neutrino	ν_τ	0	1	0	anti-tau neutrino	$\bar{\nu}_\tau$	0	-1	0

table A Leptons and anti-leptons and their properties.

PARTICLE	SYMBOL	CHARGE	LEPTON NUMBER, L	BARYON NUMBER, B	ANTI-PARTICLE	SYMBOL	CHARGE	LEPTON NUMBER, L	BARYON NUMBER, B
proton	p	+1	0	+1	anti-proton	$\bar{p}$	+1	0	-1
neutron	n	0	0	+1	anti-neutron	$\bar{n}$	0	0	-1
neutral pion	π^0	0	0	0	neutral pion	$\bar{\pi}^0$	0	0	0
pi-plus	π^+	+1	0	0	anti-pi-plus	$\bar{\pi}^+$	0	0	0
down quark	d	$-\frac{1}{3}$	0	$+\frac{1}{3}$	anti-down	$\bar{d}$	$+\frac{1}{3}$	0	$-\frac{1}{3}$
xi-minus	Ξ^-	-1	0	+1	xi-plus	Ξ^+	+1	0	-1

table B Various particles and anti-particles, and their properties.

Example: Beta-Minus (β^-) Decay

At the nucleon level, a neutron decays into a proton, an electron, and an electron antineutrino:

$$n \rightarrow p + e^- + \bar{\nu}_e$$

Let’s look at the quark transformation:

$$udd \rightarrow uud + e^- + \bar{\nu}_e \quad \text{(a down quark turns into an up quark: } d \rightarrow u + e^- + \bar{\nu}_e\text{)}$$

Let’s verify the conservation laws for this decay:

- **Charge (Q):** $0 \rightarrow (+1) + (-1) + 0 = 0$ *(Conserved)*
- **Baryon Number (B):** $(+1) \rightarrow (+1) + 0 + 0 = +1$ *(Conserved)*
- **Electron Lepton Number (L_e):** $0 \rightarrow 0 + (+1) + (-1) = 0$ *(Conserved)*

Because it involves a change in quark flavor ($d \rightarrow u$), this interaction is mediated by the **Weak Nuclear Force**.